

👍 **LevelUp Milestone** | Traffic Collisions in California

INTRODUCTION: Data is often stored across multiple tables to keep the storage requirements compact, and to organize different types of data. Knowing how to use a join is a vital skill when working with data, since bringing tables together can open the door to additional insights that are cumbersome or impossible looking at just one table at a time.

In this Milestone, you'll use your proficiency with joins to help a reporter in California use data to support an article they're writing on the causes of motor vehicle accidents. In particular, they want some information about how many accidents are caused by the influence of alcohol, or due to inattention (such as using a cell phone to text or talk to others), and when these types of accidents tend to occur.

HOW IT WORKS: Follow the prompts in the questions below to investigate your data. Post your answers in the provided boxes: the **yellow boxes** for the queries you write, **purple boxes** for visualizations and **blue boxes** for text-based answers. When you're done, export your document as a pdf file and submit it on the Milestone page – see instructions for creating a PDF at the end of the Milestone. Please don't ever remove (paste your query below 🖱️) or (write your **answer** below 🖱️). These help your Evaluator!

RESOURCES: If you need hints on the Milestone or are feeling stuck, there are multiple ways of getting help. Attend Drop-In Hours to work on these problems with your peers, or reach out to the HelpHub if you have questions. Good luck!

– Data Set **Description**

Data for this Milestone comes from the California Highway Patrol's Statewide Integrated Traffic Records System (SWITRS). The SWITRS data we've provided

(`switrs.*`) consists of two tables from the 2019 data collection: `collisions` and `parties`. The tables are related hierarchically. At the top level, there is a unique row and identifier for each incident in the `collisions` table. Then, in the lower level, each collision is between one or more parties, which include vehicles, pedestrians, etc.

The original `collisions` table has 469 664 rows and 76 columns, but we'll be focusing on only the following four columns in this Milestone:

- **case_id** - unique identifier for each collision
- **collision_time** - time of day when collision occurred, in 24 hour format
- **day_of_week** - day of week when collision occurred. Note that numbering starts at 1 = Monday and ends at 7 = Sunday (instead of 0 = Sunday)
- **party_count** - number of parties involved in the collision

The original `parties` table has 940 216 rows and 33 columns, with the following five columns of interest:

- **case_id** - associated with a collision with matching `case_id`, may not be unique
- **party_number** - numbering of parties involved, always starts from 1 for each collision
- **at_fault** - Y/N indicating whether party was at fault for collision
- **party_sobriety** - encodings for whether or not the party had been drinking
- **oaf_1**, **oaf_2** - encodings for other associated factors

Most of the features in the dataset are coded in some way for efficient data storage, which can make working with highly detailed data like this tricky. This includes the `party_sobriety`, `oaf_1`, and `oaf_2` columns you'll be investigating in the Milestone. Don't sweat that point, though: the instructions will explain the encoding values relevant to the tasks.

– **Task 1:** How frequently does alcohol use or lack of attention feature in accidents?

To start, we should run some queries on the `parties` table to understand how fault, alcohol use, and inattention are attributed to accidents.

- A.** Write a query that answers the following question: According to this dataset, how many people are at fault for a collision?

(paste your query below 📌)

```
SELECT
  COUNT(*) AS ppl_at_fault
FROM
  switrs.parties
WHERE
  at_fault = 'Y'
```

(write your **answer** below 📌)

There are 438491 people determined to be at fault.

- B.** The `party_sobriety` field takes on a value of 'B' when the party is known to have been drinking, and under the influence of alcohol. Modify your query from part A to answer the following question: How many parties were found at fault while under the influence of alcohol?

(paste your query below 📌)

```
SELECT
  case_id,
  party_number,
  at_fault,
  party_sobriety,
  oaf_1,
  oaf_2
FROM
  switrs.parties
```

```
WHERE
  at_fault = 'Y'
  AND party_sobriety = 'B'(paste your query here)
```

(write your **answer** below 📌)

There were 33512 drivers found at fault while under the influence.

- C. The **oaf_1** or **oaf_2** feature takes on a value of 'F' if inattention was a factor in the collision. Modify your query to answer the following question: How many parties were found at fault while lack of attention was a factor in the collision?

(paste your query below 📌)

```
SELECT
  case_id,
  party_number,
  at_fault,
  party_sobriety,
  oaf_1,
  oaf_2
FROM
  switrs.parties
WHERE
  at_fault = 'Y'
  AND (oaf_1 = 'F' OR oaf_2 = 'F')
```

(write your **answer** below 📌)

There were 18311 parties found at fault due to lack of attention.

– **Task 2:** When do accidents occur by day of the week?

Now that we have a way to identify whether or not a collision can be attributed to alcohol or inattention, let's add in the `collisions` table to answer the journalist's question of whether or not there are differences between the two accident sources.

- A.** Let's start with the `collisions` table on its own. Write a query that returns the number of collisions, grouped by day of the week. Which days have the highest number of collisions, and which days have the least number? Note: Day of week is encoded slightly differently than what comes out of the `date_part` function: Sunday is indicated by a 7 instead of a 0.

(paste your query below 🖱)

```
SELECT
  day_of_week,
  COUNT(*) AS total_collisions
FROM
  switrs.collisions
GROUP BY day_of_week
ORDER BY total_collisions DESC
```

(write your **answer** below 🖱)

Friday (5) has the most collisions out of all days of the week.

- B.** The `collisions` table and `parties` tables share values in the `case_id` column. Write a new query that inner joins the two tables on that column, returning the number of rows. How many rows are in the combined output table, and why?

(paste your query below 🖱)

```
SELECT
  COUNT(*) AS total_rows
FROM
```

```
switrs.collisions c
JOIN switrs.parties p ON c.case_id = p.case_id
```

(write your **answer** below 📌)

The original query stopped counting past 150000, but when counting like this, there are 940216 total rows.

- C. Combine the queries from parts A and B to return the number of collisions grouped by the day of the week. Add a condition for the involved parties so that we only count accidents where the party was found to be at fault AND under the influence of alcohol. Which days have the highest number of collisions, and which days have the smallest number?

(paste your query below 📌)

```
SELECT
  c.day_of_week,
  COUNT(*) AS total_collisions
FROM
  switrs.collisions c
  JOIN switrs.parties p
    ON c.case_id = p.case_id
WHERE
  at_fault = 'Y'
  AND party_sobriety = 'B'
GROUP BY
  day_of_week
ORDER BY
  total_collisions DESC
```

(write your **answer** below 📌)

Sunday (7) has the highest number of collisions under this criteria with 7603, and Tuesday (2) has the lowest with 3070.

- D. Modify your query to look at the number of accidents by the day of the week where the party was found to be at fault AND inattention was a factor. Which days have the highest number of collisions, and which days have the smallest number?



Try this prompt: I'm writing a SQL query that filters for accidents where the party was at fault AND inattention was a factor, using either oaf_1 = 'F' or oaf_2 = 'F'. How should I structure the WHERE clause so the OR condition is grouped correctly? What happens if I leave out the parentheses?

(paste your query below 📌)

```
SELECT
  c.day_of_week,
  COUNT(*) AS total_collisions
FROM
  switrs.collisions c
  JOIN switrs.parties p
    ON c.case_id = p.case_id
WHERE
  at_fault = 'Y'
  AND (oaf_1 = 'F' OR oaf_2 = 'F')
GROUP BY
  day_of_week
ORDER BY
  total_collisions DESC
```

(write your **answer** below 📌)

Friday and Thursday have the 2 highest counts (3030 and 2762), while Sunday and Saturday have the 2 lowest (2060 and 2273).

– Task 3: Follow-Up Analysis on Traffic Collisions

Below are three follow-up questions from the journalist. Answer each one in a way that is clear, data-backed, and useful for their article. *Remember, you can continue to use ChatGPT as your teammate in helping you answer these!*

- A. Your analysis shows that alcohol-related crashes peak on weekends, while inattention-related crashes follow the overall weekly trend. Why do you think this difference exists? What real-world factors might explain this pattern?

(write your **answer** below 🖊)

People are more likely to be partying and drinking on the weekends. That's when people are most likely to be off work. Inattention is a more common and consistent issue.

- B. If a policymaker wanted to reduce accidents caused by alcohol or inattention, what kind of interventions or regulations might be most effective based on this data?

(write your **answer** below 🖊)

You could make alcohol as hard as possible to obtain by putting time and day restrictions on when people can purchase it. For instance, stores can't sell alcohol past 6 pm on the weekends.

- C. If I wanted to dig deeper into this story, what other questions should I ask about the data?

(write your **answer** below 📌)

Do we have access to the time of day when these accidents are occurring? Do we know where these people are going and where they are coming from?

– LevelUp

Simply because an accident was such that inattention was a factor does not necessarily mean that a cell phone was the source of the driver's distraction. In the `parties` table, there is a column called `sp_info_2`. This feature takes on a value of B, 1, or 2 if a cell phone was known to be in use at the time of the accident.

- A. If you're interested in digging deeper, you might want to try seeing what proportion of accidents were caused by cell phone distraction, and if they differ from other 'inattention' accidents.

Keep in mind that the `sp_info_2` column is a string data type, so you'll need to treat the '1', and '2' codes appropriately!

(paste your query below 📌)

```
SELECT
  c.day_of_week,
  COUNT(*) AS total_collisions
FROM
  switrs.collisions c
  JOIN switrs.parties p
    ON c.case_id = p.case_id
WHERE
  at_fault = 'Y'
  AND (sp_info_2 = 'B' OR sp_info_2 = '1' OR sp_info_2 = '2')
GROUP BY
  day_of_week
```

```
ORDER BY  
total_collisions DESC
```

(write your **answer** below 🖐)

The numbers are much lower than the distracted driving query.
The day rankings have remained pretty similar with Friday being at the top and Sunday still being at the bottom.

– Submission

Great work completing this Milestone! To submit your completed Milestone, you will need to download / export this document as a PDF and then upload it to the Milestone submission page. You can find the option to download as a PDF from the File menu in the upper-left corner of the Google Doc interface.